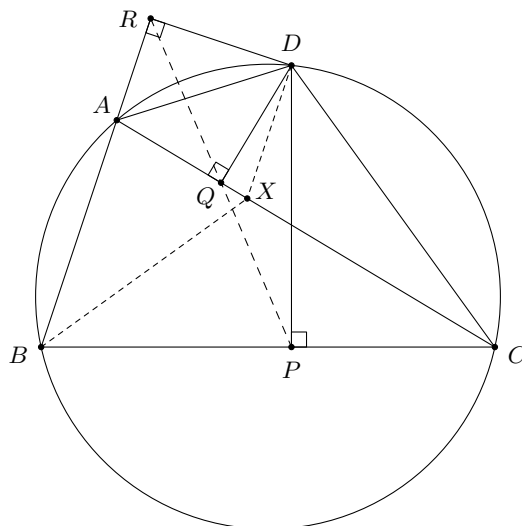


**Problem 1.** Let  $ABCD$  be a cyclic quadrilateral. Let  $P, Q, R$  be the feet of the perpendiculars from  $D$  to the lines  $BC, CA, AB$ , respectively. Show that  $PQ = QR$  if and only if the bisectors of  $\angle ABC$  and  $\angle ADC$  are concurrent with  $AC$ .



**Solution.** Firstly, let the bisector of  $\angle ABC$  meet  $AC$  at  $X_1$ , and let the bisector of  $\angle ADC$  meet  $AC$  at  $X_2$ . Then  $X_1 = X_2$  if and only if  $\frac{AX_1}{X_1C} = \frac{AX_2}{X_2C}$ . Using the angle bisector theorem, this is equivalent to

$$\frac{AB}{BC} = \frac{AD}{DC}.$$

Secondly, as  $\angle ARD = \angle DQA = 90^\circ$ , the points  $A, Q, D, R$  lie on the circle with diameter  $AD$ . By the extended sine law, we have  $\frac{QR}{\sin \angle QAR} = AD$ , which gives  $QR = AD \sin \angle BAC$ . Similarly, since  $C, D, Q, P$  are concyclic, we have  $PQ = CD \sin \angle ACB$ . Combining these, we get

$$\frac{QR}{PQ} = \frac{AD}{CD} \times \frac{\sin \angle BAC}{\sin \angle ACB} = \frac{AD}{CD} \times \frac{BC}{AB}.$$

So  $PQ = QR$  if and only if  $\frac{AB}{BC} = \frac{AD}{DC}$ , which is equivalent to the concurrency of the angle bisectors of  $\angle ABC$  and  $\angle ADC$  with the line  $AC$ .  $\square$

**Problem 2.** Find all surjective functions  $f: \mathbb{N} \rightarrow \mathbb{N}$  such that for every  $m, n \in \mathbb{N}$  and every prime  $p$ , the number  $f(m+n)$  is divisible by  $p$  if and only if  $f(m) + f(n)$  is divisible by  $p$ . (Here  $\mathbb{N}$  denotes the set of all positive integers.)

**Solution.** Since  $f$  is surjective, there exists  $k \in \mathbb{N}$  such that  $f(k) = 1$ . If  $k \geq 2$ , we put  $m = 1$  and  $n = k - 1$  so that

$$p \mid f(k) = 1 \quad \Leftrightarrow \quad p \mid f(1) + f(k-1).$$

Since  $p \nmid 1$  for any prime  $p$ , we have  $p \nmid f(1) + f(k-1)$  for any prime  $p$ . But this is impossible since  $f(1) + f(k-1)$  is an integer larger than 1. So  $k = 1$  and  $f(1) = 1$ .

Putting  $n = 1$ , we get

$$p \mid f(m+1) \quad \Leftrightarrow \quad p \mid f(m) + 1.$$

In particular, this implies  $f(m)$  and  $f(m+1)$  are relatively prime for any  $m \in \mathbb{N}$ . Indeed, if there exists a prime dividing both of them, then the above assertion implies  $p \mid f(m) + 1$  and hence  $p \mid 1$ . Replacing  $n$  by  $n+1$ , we have

$$p \mid f(m+n+1) \quad \Leftrightarrow \quad p \mid f(m) + f(n+1). \quad (1)$$

For each fixed  $n \in \mathbb{N}$ , we claim that  $f(n+1) - f(n) = \pm 1$ . Suppose on the contrary that  $p \mid f(n+1) - f(n)$  for some prime  $p$ . Since  $f$  is surjective, we can choose  $m \in \mathbb{N}$  such that  $f(m) \equiv -f(n) \pmod{p}$ . For this choice of  $m$ , we have  $p \mid f(n+1) + f(m)$ . Using (1), we get  $p \mid f(m+n+1)$ . As  $f(m+n+1)$  and  $f(m+n)$  are relatively prime, we must have  $p \nmid f(m+n)$ . However, from the given assertion

$$p \mid f(m+n) \Leftrightarrow p \mid f(m) + f(n),$$

since  $p \mid f(m) + f(n)$  from the choice of  $m$ , we also obtain  $p \mid f(m+n)$ . This is a contradiction. Therefore,  $f(n+1) - f(n)$  has no prime divisor, and we must have

$$f(n+1) = f(n) \pm 1.$$

We now claim that  $f$  is injective. Assume  $f(a) = f(b)$  for some  $a > b$ . Note that we have

$$p \mid f(m+a) \Leftrightarrow p \mid f(m) + f(a)$$

and

$$p \mid f(m+b) \Leftrightarrow p \mid f(m) + f(b).$$

The right-hand sides of the two assertions are equivalent. So

$$p \mid f(m+a) \Leftrightarrow p \mid f(m+b) \tag{2}$$

for any  $m \in \mathbb{N}$ . For each  $n \in \mathbb{N}$ , let  $n \equiv r \pmod{a-b}$  where  $r = 1, 2, \dots, a-b$ . Then using (2) repeatedly, we get

$$p \mid f(n) \Leftrightarrow p \mid f(n - (a-b)) \Leftrightarrow p \mid f(n - 2(a-b)) \Leftrightarrow \dots \Leftrightarrow p \mid f(r).$$

So the prime divisors of  $f(n)$  can only be the prime divisors of  $f(r)$ . Since there are only finitely many choices for  $r$ , we can pick a prime  $p$  such that  $p \nmid f(r)$  for  $r = 1, 2, \dots, a-b$ . But there exists  $n \in \mathbb{N}$  such that  $f(n) = p$  from the surjectivity. This is a contradiction since the prime divisor of  $f(n)$  is distinct from those of the  $f(r)$ 's. So  $f$  is injective.

Starting from  $f(1) = 1$ , we can use  $f(n+1) = f(n) \pm 1$  to prove inductively that  $f(n) = n$  for any  $n \in \mathbb{N}$ . Indeed, if there is a smallest  $k$  for which  $f(k) \neq k$ , then we must have  $f(k) = f(k-1) - 1 = k-2 = f(k-2)$ . This contradicts the injectivity of  $f$ . It is easy to check that  $f(n) = n$  is the only solution.  $\square$